A QUASI-EXCELLENT RING IS NAGATA

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finite type algebras over a quasi-excellent ring are quasi-excellent. Lemma \ref{lemma-finite-type-over-excellent}, cited at proof lines 13469-13470. Reduces the claim to quasi-excellent domains.
- The criterion for the universally Japanese property, and its characterisation through J-2 and the local case. Algebra, Lemma \ref{algebra-lemma-check-universally-japanese} and Algebra, Lemma \ref{algebra-lemma-characterize-N-1}, cited at proof lines 13470-13474. Reduce to quasi-excellent local domains.
- The completion of a quasi-excellent local ring is reduced, i.e. the ring is analytically unramified, and analytic unramifiedness implies the universal Japanese property. Lemma \ref{lemma-reduced-goes-up} and Algebra, Lemma \ref{algebra-lemma-analytically-unramified-easy}, cited at proof lines 13476-13482. Give the final step.

Source reference. Original lemma 07QV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9845–9845. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07QU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9844–9844. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0351: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4112–4112. Navigation pointer to a statement the

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Source reference. Original lemma 0333: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4042–4042. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07QK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9835–9835. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 032Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4037–4037. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *A quasi-excellent ring is Nagata.*

Proof. Let R be quasi-excellent. Using that a finite type algebra over R is quasi-excellent (Lemma 07QU) we see that it suffices to show that any quasi-excellent domain is N-1, see Algebra, Lemma 0351. Applying Algebra, Lemma 0333 (and using that a quasi-excellent ring is J-2) we reduce to showing that a quasi-excellent local domain R is N-1. As $R \to R^\wedge$ is regular we see that $R^\wedge$ is reduced by Lemma 07QK. In other words, R is analytically unramified. Hence R is N-1 by Algebra, Lemma 032Y. $\square$